

SURAJ RAVINDRA KAPARE

Data Scientist

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PROFESSIONAL SUMMARY

M.S. in Data Science with experience developing predictive, NLP, and computer vision solutions across datasets ranging from 800 images to 129K+ records. Built a LightGBM passenger-satisfaction model achieving 96% accuracy and developed a GWU Capstone combining Claude, SAM, and regression analysis to identify interpretable predictors of urban aesthetic ratings. Additional experience includes deploying an AWS-based NLP pipeline using S3, Lambda, SageMaker, and API Gateway.

PROFESSIONAL EXPERIENCE

Data Analyst Intern | ETLHive Jan 2024 – May 2024
Customer Shopping Behavior Analytics Pune, India

- Built the end-to-end analysis of 3,900 customer-shopping records, using Python/Pandas to clean, transform, and engineer customer features before loading analysis-ready data into PostgreSQL.
- Developed PostgreSQL analyses using CTEs, subqueries, CASE statements, and window functions to segment customers and uncover purchasing patterns, including 2,518 repeat buyers who remained non-subscribers—a key subscription-growth opportunity.
- Delivered an interactive Power BI dashboard translating customer, revenue, product, discount, and subscription trends into recommendations for targeted marketing, loyalty programs, subscription adoption, and product positioning.

PROJECT EXPERIENCE

Beyond Black-Box Assessment: Visual Predictors of Urban Aesthetic Attraction Jan 2026 – May 2026
GWU Capstone

- Developed a theory-driven framework using Claude (LLM scoring) and SAM to evaluate 800 urban images across Alexander's 15 properties of wholeness, applying regression analysis to identify interpretable predictors of human aesthetic ratings – directly addressing NIMA's under-discrimination (std dev: 0.246 vs. human 1.598).

Airline Passenger Satisfaction Analysis & Prediction | GWU Sep 2024 – Dec 2024

- Analyzed 129K+ passenger survey records using Python to evaluate demographic, travel, delay, and service-quality factors, identifying inflight WiFi, online boarding, and travel class as key drivers of passenger satisfaction.
- Built and evaluated Logistic Regression and LightGBM classifiers using 5-fold cross-validation; LightGBM achieved 96% accuracy and ROC-AUC, outperforming Logistic Regression's 88% accuracy.

YouTube Comments Sentiment Analysis – AWS Cloud Pipeline Sep 2025 – Dec 2025

- Developed an end-to-end AWS NLP pipeline using S3, Lambda, and SageMaker to preprocess YouTube comments and classify sentiment as positive, neutral, or negative.
- Deployed the model through a SageMaker endpoint and built a REST API with API Gateway and Lambda to deliver real-time sentiment predictions for new comments.

EDUCATION

The George Washington University Aug 2024 – May 2026
M.S. in Data Science, GPA: 3.64 Washington, DC

Savitribai Phule Pune University Aug 2018 – Jun 2022
B.E. in Mechanical Engineering Pune, India

TECHNICAL SKILLS

Programming & Data: Python (Pandas, NumPy, Scikit-learn, NLTK), SQL, PostgreSQL, MongoDB, R, Excel, Git, GitHub

AI/ML: PyTorch, LightGBM, Logistic Regression, NLP, Feature Engineering, Model Evaluation, LLM Pipelines, Hugging Face, RAG

Analytics & Apps: Power BI, Tableau, Streamlit

Cloud & Data Platforms: AWS (S3, Lambda, SageMaker, API Gateway), ETL Pipelines, Serverless Architecture